Ishan Sinha

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Education

Michigan State University East Lansing, MI

Bachelor of Science in Computer Science | Concentration: Artificial Intelligence *Aug. 2024 – Dec. 2027*

- Minors: Data Science; Computational Math, Science & Engineering; Mathematics | **Michigander Scholar**

Experience

Auto-Owners Insurance Lansing, MI

Software Development Intern *May 2026 – Aug. 2026*

- Engineered decoupled Java Spring Boot microservices to optimize Guidewire Claim Center configurations, ensuring high availability and reducing latency for large-scale claims by 8%.
- Enhanced Python backend update workflows to aggregate relevant data, streamlining the visual presentation of internal user profiles and resolving query bottlenecks.

SEE-Insight Lab East Lansing, MI

Undergraduate Researcher *Sep. 2026 – Present*

- Architected knowledge representations and LLM prompt logic using Python for an AI Teaching Assistant, establishing quantitative evaluation rubrics to benchmark over agentic responses.
- Conducted lab-wide code reviews and triaged GitHub pull requests, auditing legacy machine learning scripts to enforce strict software engineering best practices and clean repository hygiene.

Facility for Rare Isotope Beams (FRIB) East Lansing, MI

Undergraduate Researcher *Jan. 2026 – Sep. 2026*

- Engineered a custom 3D sparse neural network framework using PyTorch and C++, accelerating rare event detection across high-dimensional physical datasets.
- Optimized low-level CUDA kernels and sparse tensor operations, significantly reducing compute latency and memory overhead for complex 3D structural analysis.

MSU Unmanned Systems East Lansing, MI

Software Team Lead (C-UASC Campaign) *Nov. 2025 – Present*

- Directed autonomous drone software execution, deploying a C++ edge-compute stack on an NVIDIA Jetson Orin Nano to actuate payloads via MAVLink telemetry.
- Engineered custom spherical math algorithms for spatial processing, resolving waypoint navigation errors down to a 0.55-meter accuracy to secure 3rd place overall.

MSU AI Club East Lansing, MI

Workshop Coordinator *Aug. 2026 – Present*

- Directed technical curriculum for 100+ members, scaling workshop topics from broad architectural discussions to hands-on machine learning implementations.
- Engineered software workarounds for constrained local environments, translating complex mathematical concepts like Bayesian probability into accessible, interactive Python models.

Projects

OpticHash | *Java (Spring Boot), C++, PyTorch, FastAPI, Docker, Nginx*

- Architected a decoupled B2B microservices network using a Java Spring Boot API Gateway and Nginx proxy to stream multi-part payloads across multi-stage Docker containers.
- Authored a native C++ microservice utilizing OpenCV to evaluate structural geometry, generating O(1) Perceptual Hashes (pHash) that shield downstream ML nodes from redundant processing.
- Compressed an edge-focused PyTorch MobileNetV3 model via INT8 dynamic quantization, slashing VRAM footprint by 75% while achieving low-latency inference profiled at 58.6M FLOPS.

NBA Finals NLP Predictor | *Python, PyTorch, Hugging Face (RoBERTa, Whisper), ChromaDB*

- Built an end-to-end RAG pipeline utilizing ChromaDB and semantic chunking to parse 1000+ media transcript snippets, achieving zero-error accuracy predicting the outright series champion.
- Processed media through a custom RoBERTa pipeline to map 7-dimensional emotional feature vectors, deploying a Random Forest architecture for low-compute local inference without overfitting.

Technical Skills

Languages: Python, Java, C/C++, C#, JavaScript, SQL, HTML/CSS
Frameworks: PyTorch, Scikit-Learn, Hugging Face (RoBERTa, Whisper), Spring Boot, FastAPI, ChromaDB, OpenCV, TensorRT, CUDA
Proficiencies: RAG Pipelines, Microservices Architecture, Compute Optimization, Unsupervised Learning
Tools: Git, GitHub, Docker, Nginx, CI/CD, MAVLink, REST APIs, Linux/Unix